1. MOMENT OF INERTIA OF A SOLID SPHERE OF KNOWN MASS USING VERNIER CALIPER

AIM

To determine the moment of inertia of a solid sphere of known mass using Vernier caliper

APPARATUS REQUIRED

Vernier caliper, Solid sphere

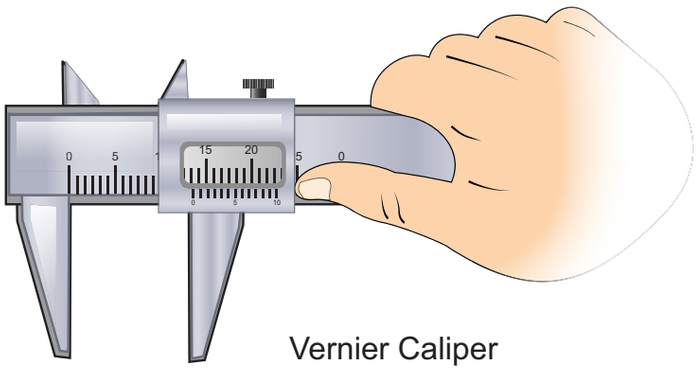
FORMULA

Moment of inertia of a solid sphere about its diameter $I_d = \frac{2}{5} MR^2 \text{ kgm}^2$

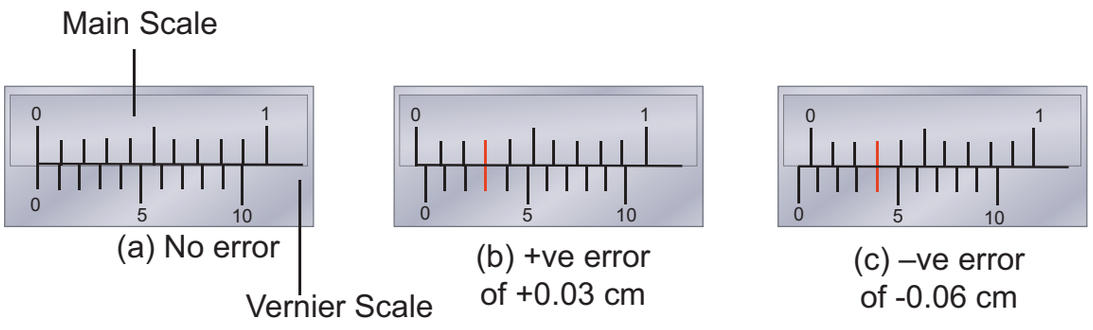
Where M ← Mass of the sphere (known value to be given) in kg

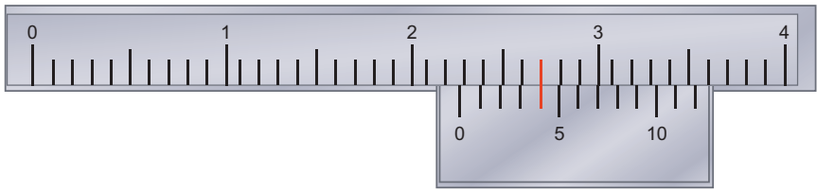
R ← Radius of the sphere in metre

DIAGRAM



Vernier Caliper





(d) Vernier reading

A model reading
MSR = 2.2 cm ; VSC = 4 divisions;
Reading = [2.2 cm+(4x0.01cm)] = 2.24 cm

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